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Semiparametric Joint Modeling for Survival Analysis With Longitudinal Covariates

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ABSTRACT

Characterizing the association between survival time and the dynamic patterns of a longitudinal covariate trajectory is of particular interest in many studies. Classical time-dependent survival models focus mainly on the link between the concurrent covariate value and the instantaneous hazard function. Consequently, the conditional survival function is often not properly defined on the whole time range, which causes difficulty in model estimation and interpretation. In this article, we propose a novel semiparametric joint modeling approach, in which the observed longitudinal trajectory is modeled as a random realization of a latent functional pattern. We assume each latent pattern uniquely indexes a global survival function via a log-linear functional regression model. Because the observational time interval of the longitudinal data depends on the survival time, we propose to jointly model the longitudinal and survival data. By using the latent pattern as an infinite-dimensional shared parameter, our approach extends the classical parametric joint modeling method to a semiparametric setting. We show that the proposed estimator achieves the semiparametric efficiency bound. Simulation studies and a real data application demonstrate the advantageous finite sample performances of our new approach. Supplementary materials for this article are available online.

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1. Introduction

A goal of survival analysis is to determine the conditional survival distribution given the observed covariate information, such as estimating the remaining survival times and predicting future survival probabilities. Since the seminal paper of Cox (1972), the association of the survival time with the covariate is predominantly formulated through the hazard-based regression models. When the covariate is time-invariant, there is a simple one-to-one correspondence between the hazard function and the survival function. When the covariate is measured longitudinally that may change over time, it is typically treated as a time-varying covariate in an instantaneous hazard model. While this approach provides flexible inference based on classical partial likelihood (Cox 1975) and NPML approaches (Breslow 1972; Andersen et al. 1993; Kalbfleisch and Prentice 2002), only defining the hazard function no longer guarantees a well-defined survival function when the covariate is an internal time-varying variable (Kalbfleisch and Prentice 2002), which is better known as the endogenous variable in joint modeling literature (Diggle et al. 2002; Rizopoulos 2012). In this article, we propose an alternative model for survival analysis with longitudinal covariates based on a new semiparametric joint modeling approach, which handles the time-varying problem under a time-invariant framework. This approach facilitates direct estimation of the conditional survival functions.

To illustrate the problem more clearly, denote by $x(t)$ the longitudinal covariate at time t . Let $\bar{X}(t) = \{x(s), 0 \leq s < t\}$ be the longitudinal trajectory. Given $\bar{X}(t)$, the conditional survival function is defined as $S(t|\bar{X}(t)) = P(T \geq t|\bar{X}(t)) = \exp \left\{ - \int_0^t \lambda(s|\bar{X}(t)) ds \right\}$, where $\lambda(s|\bar{X}(t))$ is the conditional hazard function at time s . To obtain $S(t|\bar{X}(t))$, we need not only $\lambda(t|\bar{X}(t))$ but also $\lambda(s|\bar{X}(t))$ for $s \leq t$. For exogenous covariates such as environmental factors, we can assume that $\lambda(s|\bar{X}(t)) = \lambda(s|\bar{X}(s))$. In this case, $S(t|\bar{X}(t)) = \exp \left\{ - \int_0^t \lambda(s|\bar{X}(s)) ds \right\}$ is well-defined with only the definition of $\lambda(s|\bar{X}(s))$. It implies that the occurrence of the survival event in $[s, s + \Delta s)$ does not depend on the future covariate path $x(t)$ for $t \geq s + \Delta s$. However, most longitudinal covariates are biomarkers of the event and thus endogenous. For example, in chronic kidney disease (CKD) studies where the estimated glomerular filtration rate (eGFR) is the most common longitudinal covariate, a good eGFR at a latter time would indicate a low risk of CKD prior to that time. For these endogenous covariates, $\lambda(s|\bar{X}(t)) \neq \lambda(s|\bar{X}(s))$ and $S(t|\bar{X}(t))$ is no longer properly defined by only the instantaneous hazard function (see also Kalbfleisch and Prentice 2002, chap. 6.3). When the interest is in estimation of the survival distribution and remaining survival times, alternative approaches are needed.

Our new functional approach was motivated by the observation that, in many studies, the global pattern of longitudinal

biomarker trajectory is more important than its time-varying values in determining the survival probabilities. For example, in the African American Study of Kidney Disease and Hypertension (AASK, Wright et al. 2002), researchers are interested in how the eGFR trajectory can be used to estimate the time to end-stage renal disease (ESRD). Prior work showed that many patients had nonlinear longitudinal eGFR patterns, and that these overall patterns, rather than concurrent values or instantaneous slopes, were most predictive to ESRD (Li et al. 2012). Figure 1 illustrates three such trajectories. In our real data analysis presented in Section 6, we found two key patterns that are most associated with the time to ESRD.

When the longitudinal data are known to be an endogenous covariate related to the survival time, joint modeling is a popular approach that explicitly quantifies the associations of the longitudinal and survival data, see Tsiatis and Davidian (2004) and Rizopoulos (2012) for a comprehensive review. In a classical joint modeling approach (e.g., Wu and Carroll 1988; DeGruttola and Tu 1994; Wulfsohn and Tsiatis 1997; Schluchter, Konstan, and Davis 2002; Zeng and Cai 2005; Tseng, Hsieh, and Wang 2005), a mixed-effects model is used to model the longitudinal biomarker trajectories, and the person-specific random effects from the mixed-effects model are included as time-independent covariates in a time-to-event model. The longitudinal model and the survival model are jointly estimated by integrating out the random coefficients as shared parameters. This approach was further extended to include different association structures between the outcomes (Rizopoulos et al. 2014), and multiple longitudinal biomarkers where the predictive features may vary for different biomarkers (Andrinopoulou and Rizopoulos 2016; Rizopoulos and Taylor 2024).

In this article, we adopt a similar approach as the shared-parameter models and link the longitudinal model and survival model by a latent functional pattern defined in a common time domain. We assume the longitudinal data are realizations of this latent functional pattern and observed in a subinterval determined by the follow-up time. The survival time is assumed to

follow a log-linear model with the latent pattern as a functional predictor. The latent pattern is then used as the functional shared parameter of the longitudinal and survival models. In contrast to the traditional parametric joint models where the shared parameters are defined in a finite dimensional space, the latent pattern is a functional shared parameter which takes value in an infinite dimensional space. To implement our model, we assume the latent functional pattern follows a Gaussian process. Because each realization of Gaussian process is a curve (Wahba 1978), we approximate the pattern using a B-spline basis expansion. The number of knots is allowed to go to infinity as the number of observations and per subject and number of subjects go to infinity, which extends the classical B-spline methods (Rice and Wu 2001) to a nonparametric setting. For the survival part, the log-hazard function of the residual term is modeled by another B-spline expansion, which was previously studied by Ding and Nan (2011) for an AFT model with baseline covariates.

In the shared-parameter approach, several authors have also used parametric B-spline expansions to flexibly model the longitudinal trajectories (e.g., Brown, Ibrahim, and DeGruttola 2005; Li, Xiao, and Luo 2022) and used the random B-splines coefficients as shared parameters. However, the B-spline coefficients by themselves do not have clear interpretation in contrast to our functional shared parameters that can be interpreted as the latent trajectories. To compute the joint likelihood, one has to integrate out the random effects. This integration is high dimensional in the approach of linking the joint model by the random B-spline coefficients, and prevents the number of basis functions to increase with the sample size in a semiparametric setting. To alleviate the computational burden, Rizopoulos, Verbeke, and Lesaffre (2009) proposed a Laplace approximation to compute the high dimensional integration and in the setting with multiple biomarkers; De Witte et al. (2025) proposed a pairwise fitting algorithm. In contrast, our approach treats the latent random function as a basis unit and the resultant integration is one-dimensional. This not only leads to a manageable computational

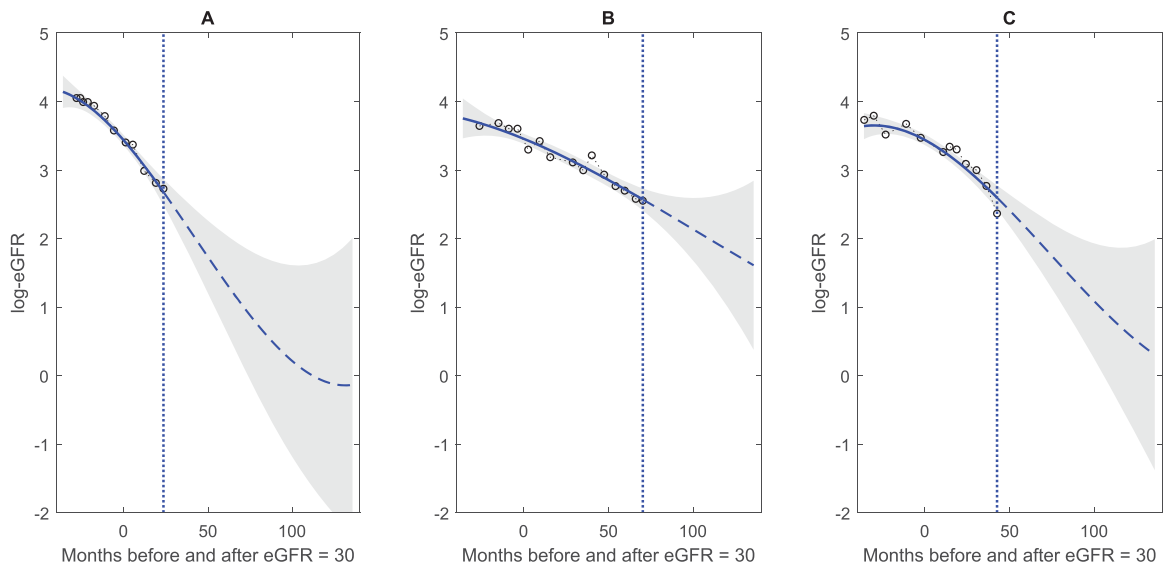


Figure 1. Three longitudinal eGFR trajectories in AASK. Circles: the observed data; solid line: the fitted curve within observation window; solid+dashed line: the fitted latent pattern; shaded area: the 95% confidence band ; vertical line: the onset time of ESRD.

task, but also enables investigation of the related asymptotic properties of our semiparametric estimators.

When the functional data are fully observed at baseline without measurement error, several authors have proposed to incorporate the functional regression into survival analysis (Gellar et al. 2015; Ye, Li, and Guan 2015; Qu, Wang, and Wang 2016; Kong et al. 2018; Liu et al. 2025). Their methods are not applicable to longitudinal trajectories sparsely observed in informative windows. The key difficulty of applying the functional method to this problem is that the observational time window is usually determined by the length of the survival time. Therefore, joint modeling is a necessary approach to consider the longitudinal patterns and the survival data simultaneously.

Theoretically, the proposed survival submodel can be viewed as a functional regression model where the functional covariate is sparsely measured with noises in an informative time interval. No existing theory is readily available for this problem, because by joint modeling we need to estimate the latent functional covariate and the survival model simultaneously. The overall convergence rate of all the estimated parameters will be a slower two-dimensional rate dictated by the covariance kernel function of the latent functional pattern. In this article, building on the sieve theories developed in Shen and Wong (1994) and Ding and Nan (2011), we prove not only the semi-parametric efficiency of our model but also the optimal convergence rate for the one-dimensional functional estimates.

2. Model

2.1. Semiparametric Joint Model

Suppose the study has n subjects indexed by $i = 1, 2, \dots, n$. For the i th subject, let T_i be the survival time, C_i be the right-censoring time, and $\mathbf{z}_i = (z_{i1}, \dots, z_{id})^\top$ be a d -dimensional vector of time-independent covariates. Meanwhile, suppose that we observe a covariate trajectory $y_i(t_{ij})$, $j = 1, \dots, m_i$, where t_{ij} denotes the measurement time. Define $\mathbf{y}_i = (y_i(t_{i1}), \dots, y_i(t_{im_i}))^\top$. We define a conceptual common domain $\mathcal{T} = [0, \tau]$, such that $\mathbf{t}_i = (t_{i1}, \dots, t_{im_i}) \subset \mathcal{T}$ for $i = 1, \dots, n$.

We assume that the longitudinal trajectory y_i is a random realization of a population profile $\mu(t)$ and a latent curve $x_i(t)$ defined on $\mathcal{T}$, which is observed with noises on $\mathbf{t}_i$. The latent pattern $x_i(t)$ is linked to the survival time T_i by a semiparametric functional linear model. Then our model can be written as

$$\begin{aligned} y_{ij} &= \mu(t_{ij}) + x_i(t_{ij}) + e_{ij}, \quad i = 1, \dots, n, \quad j = 1, \dots, m_i, \\ \log T_i &= \boldsymbol{\xi}^\top \mathbf{z}_i + \int_{\mathcal{T}} \beta(t) x_i(t) dt + \varepsilon_i, \end{aligned} \quad (2)$$

where $e_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma_e^2)$ are measurement errors, $\beta(t)$ is a smooth coefficient function, $\boldsymbol{\xi} = (\xi_1, \dots, \xi_d)^\top$ is the coefficients of $\mathbf{z}_i$, and ε_i is the residual term with a common but unspecified distribution. The random curve $\{x_i(t), t \in \mathcal{T}\}$ is defined as the *latent pattern* of the biomarker trajectory $\mathbf{y}_i$. We assume that $x_i(t)$ follows a Gaussian process with mean 0 and covariance function $\text{cov}(x_i(s), x_i(t)) = R(s, t)$ for $s, t \in \mathcal{T}$. The conceptual common domain $\mathcal{T}$ is a mathematical framework to facilitate comparable latent functional patterns $x_i(\cdot)$ across different subjects, which are used to index the conditional distribution of the survival

time $f(T_i | x_i(\cdot))$. The latent pattern $x_i(\cdot)$ is estimated based on the observed trajectory $\mathbf{y}_i$ and integrated out in the joint estimation, providing a link between the longitudinal data and the survival time. In some special cases, the latent functional pattern $x_i(\cdot)$ can be interpreted as the biomarker trajectory if a subject were to survive to τ , the maximal follow-up time. However, in general, we do not require $x_i(\cdot)$ to physically exist in the entire interval $\mathcal{T}$ as long as it is estimable given the observed data $\mathbf{y}_i$.

Remark 1. The functional pattern $x_i(\cdot)$ can be treated as a time-independent feature extracted from the longitudinal data. Transforming a time-varying longitudinal covariate into time-independent features facilitates a one-to-one correspondence between the survival and hazard functions, which is no longer maintained in a time-varying hazard model.

Remark 2. In the semiparametric joint model (1) and (2), we assume the functional curve $x_i(\cdot)$ is observed on the same time domain as the survival time that are subject to right-censoring. In this case, joint modeling is necessary to handle the inherent correlation between the longitudinal and survival data. This is different from the existing functional survival models, for example, functional Cox model (Qu, Wang, and Wang 2016) and functional AFT model (Liu et al. 2025), where the functional data are fully observed at baseline without censoring.

2.2. Conditional Distribution of the Survival Time

By the definition of $x_i(t)$ and y_{ij} , we obtain that $x_i(t)$ conditional on the longitudinal trajectory $\mathbf{y}_i$ follows a Gaussian process with conditional mean $E\{x_i(t) | \mathbf{y}_i\} = R(t, \mathbf{t}_i)(R(\mathbf{t}_i, \mathbf{t}_i) + \sigma_e^2 I_{m_i})^{-1} \mathbf{y}_i$, and conditional covariance function $\text{cov}\{x_i(t), x_i(s) | \mathbf{y}_i\} = R(t, s) - R(t, \mathbf{t}_i)(R(\mathbf{t}_i, \mathbf{t}_i) + \sigma_e^2 I_{m_i})^{-1} R(\mathbf{t}_i, s)$, where $R(\mathbf{t}, \mathbf{s}) = [R(t_k, s_l)]_{k=1, \dots, K; l=1, \dots, L}$ is a $K \times L$ matrix for any $\mathbf{t} = (t_1, \dots, t_K)$ and $\mathbf{s} = (s_1, \dots, s_L)$. Assume $R(t, s)$ is continuous and square integrable. The above Gaussian process admits a Gaussian measure $\mathcal{F}(x_i | \mathbf{y}_i)$ (Rajput and Cambanis 1972), which is the conditional probability measure of the latent pattern $x_i(\cdot)$ given $\mathbf{y}_i$.

Let λ and Λ respectively denote the hazard rate and the cumulative hazard function of ε_i , that is, the residual term in model (2). The conditional survival function of the T_i given $(\mathbf{z}_i, x_i)$ is defined as

$$\begin{aligned} S_T(s | \mathbf{z}_i, x_i) &= \Pr(T_i > s | \mathbf{z}_i, x_i) \\ &= \Pr\left(\varepsilon_i > \log s - \boldsymbol{\xi}^\top \mathbf{z}_i - \int_{\mathcal{T}} \beta(t) x_i(t) dt\right) \\ &= \exp\left\{-\Lambda\left(\log s - \boldsymbol{\xi}^\top \mathbf{z}_i - \int_{\mathcal{T}} \beta(t) x_i(t) dt\right)\right\}. \end{aligned}$$

By the law of conditional probability, we can derive the conditional survival function of T_i given the observed covariate data $(\mathbf{z}_i, \mathbf{y}_i)$ as

$$\begin{aligned} S_T(s | \mathbf{z}_i, \mathbf{y}_i) &= \Pr(T_i > s | \mathbf{z}_i, \mathbf{y}_i) = E_{x_i}[\Pr(T_i > s | \mathbf{z}_i, x_i) | \mathbf{y}_i] \\ &= E_{x_i}[S_T(s | \mathbf{z}_i, x_i) | \mathbf{y}_i]. \end{aligned}$$

Let $w_i = \int_{\mathcal{T}} \beta(t) x_i(t) dt$. Because $S_T(s | \mathbf{z}_i, x_i)$ depends on $x_i(\cdot)$ only through w_i , the above conditional survival function can be reduced to require only a one-dimensional integral:

$$S_T(s|\mathbf{z}_i, \mathbf{y}_i) = \int_{w_i} \exp \left\{ -\Lambda \left(\log s - \boldsymbol{\xi}^\top \mathbf{z}_i - w_i \right) \right\} f_w(w_i|\mathbf{y}_i) dw_i,$$

where $f_w(w_i|\mathbf{y}_i)$ is the Gaussian density with mean $\int_{\mathcal{T}} \beta(t) E\{x_i(t)|\mathbf{y}_i\} dt$ and variance $\int_{\mathcal{T}} \int_{\mathcal{T}} \beta(v) \text{cov}[x_i(v), x_i(t)|\mathbf{y}_i] \beta(t) dv dt$.

The term w_i is a time-independent index that allows an explicit modeling of the conditional survival function in a similar way as the classical time-independent models. It is estimated by the observed longitudinal trajectory $\mathbf{y}_i$ via the conditional distribution $f_w(w_i|\mathbf{y}_i)$, which incorporates the time-varying information into the survival model. Since $f_w(w_i|\mathbf{y}_i)$ is completely determined by the observed data, no explicit extrapolation is needed to estimate the conditional survival function. The individual mean survival time can be obtained by $E\{T_i|\mathbf{z}_i, \mathbf{y}_i\} = \int_0^\infty S(t|\mathbf{z}_i, \mathbf{y}_i) dt$. We can define the individual restricted mean survival time (RMST, see, e.g., Tian, Zhao, and Wei 2013) as $R\{T_i|\mathbf{z}_i, \mathbf{y}_i\} = \int_0^{T_M} S(t|\mathbf{z}_i, \mathbf{y}_i) dt$.

Remark 3. In the context of dynamic predictions (Rizopoulos 2011), the i th subject is typically event free up to the last follow-up time t_{i,m_i} . By Bayesian Theorem, the conditional survival function can be estimated as

$$S(t|T_i > t_{i,m_i}, \mathbf{z}_i, \mathbf{y}_i) = \frac{\Pr(T_i > t|\mathbf{z}_i, \mathbf{y}_i)}{\Pr(T_i > t_{i,m_i}|\mathbf{z}_i, \mathbf{y}_i)} = \frac{S_T(t|\mathbf{z}_i, \mathbf{y}_i)}{S_T(t_{i,m_i}|\mathbf{z}_i, \mathbf{y}_i)},$$

for $t > t_{i,m_i}$. This can be easily updated if we observe a new longitudinal data y_{i,m_i+1} at the next time point t_{i,m_i+1} , by replacing the numerator and denominator with $S_T(t|\mathbf{z}_i, \mathbf{y}_i^*)$ and $S_T(t_{i,m_i+1}|\mathbf{z}_i, \mathbf{y}_i^*)$, where $\mathbf{y}_i^* = (\mathbf{y}_i, y_{i,m_i+1})$ denotes the updated longitudinal data.

Given the conditional survival function, we can also estimate the remaining survival time after t_{i,m_i} , defined as, $E\{T_i - t_{i,m_i}|T_i > t_{i,m_i}, \mathbf{z}_i, \mathbf{y}_i\} = \int_{t_{i,m_i}}^\infty S(t|T_i > t_{i,m_i}, \mathbf{z}_i, \mathbf{y}_i) dt$. Similar to RMST, we can have a restricted version of the remaining survival time as $R\{T_i - t_{i,m_i}|T_i > t_{i,m_i}, \mathbf{z}_i, \mathbf{y}_i\} = \int_{t_{i,m_i}}^{T_M} S(t|T_i > t_{i,m_i}, \mathbf{z}_i, \mathbf{y}_i) dt$. The properties of dynamic prediction under our new model are beyond the scope of this article and deserve careful further studies.

3. Estimation

3.1. Joint Likelihood

Due to right censoring, the observed data are $\{(S_i, \delta_i, \mathbf{z}_i, \mathbf{y}_i)\}$, where $S_i = \min\{T_i, C_i\}$, $\delta_i = I(T_i \leq C_i)$. To estimate the proposed model, we construct the joint distribution of the longitudinal and survival time data based on models (1) and (2). The parameters of interest are summarized as $\boldsymbol{\theta} = (\boldsymbol{\xi}, \beta(\cdot), \lambda(\cdot), R(\cdot, \cdot), \mu(\cdot), \sigma_e^2)$. For the longitudinal submodel (1), we assume that the measuring schedule $\mathbf{t}_i$ is noninformative, which is also independent of the random curves $x_i(\cdot)$. Moreover, the censoring time C_i is assumed to be independent of $(T_i, x_i(\cdot))$ conditional on $\mathbf{z}_i$.

For simplicity of notation, we define $r_i = \log S_i = \min\{\log T_i, \log C_i\}$. Then the i th observed data can also be denoted by $(r_i, \delta_i, \mathbf{z}_i, \mathbf{y}_i)$. It is easy to see that $(r_i, \delta_i, \mathbf{z}_i)$ and $\mathbf{y}_i$

are independent conditional on $x_i(\cdot)$. Thus, the joint density function can be written as

$$f(r_i, \delta_i, \mathbf{z}_i, \mathbf{y}_i; \boldsymbol{\theta}) = \int_{x_i \in \mathcal{X}(\mathcal{T})} f_r(r_i, \delta_i|\mathbf{z}_i, x_i; \boldsymbol{\xi}, \beta, \lambda) \times f_y(\mathbf{y}_i|x_i; R, \mu, \sigma_e^2) d\mathcal{F}(x_i; R) \times H_i, \quad (3)$$

where $\mathcal{X}(\mathcal{T})$ is the separable Hilbert space characterized by the covariance operator R , and $\mathcal{F}(x_i; R)$ is the Gaussian measure defined on $\mathcal{X}(\mathcal{T})$ that corresponds to the Gaussian process with mean 0 and covariance kernel $R(\cdot, \cdot)$ (Rajput and Cambanis 1972). The conditional density function f_r for the survival data (r_i, δ_i) is given as

$$f_r(r_i, \delta_i|\mathbf{z}_i, x_i; \boldsymbol{\xi}, \beta, \lambda) = \lambda \left(r_i - \boldsymbol{\xi}^\top \mathbf{z}_i - \int_{\mathcal{T}} \beta(t) x_i(t) dt \right)^{\delta_i} \times \exp \left\{ -\Lambda \left(r_i - \boldsymbol{\xi}^\top \mathbf{z}_i - \int_{\mathcal{T}} \beta(t) x_i(t) dt \right) \right\}.$$

The function $f_y(\mathbf{y}_i|x_i; R, \mu, \sigma_e^2)$ is the Gaussian density with mean $\mu(\mathbf{t}_i) + x_i(\mathbf{t}_i)$ and covariance $\sigma_e^2 \mathbf{I}_{m_i}$. H_i only depends on the conditional density distribution of C_i given $\mathbf{z}_i$ and the marginal distribution of $(\mathbf{z}_i, \mathbf{t}_i)$, and is free of $\boldsymbol{\theta}$. Without loss of generality, we ignore H_i in the following discussions for simplicity of notation. It is worth mentioning that $\mathcal{F}(x_i; R)$ does not have a well-defined Radon-Nikodym derivative w.r.t. the Lebesgue measure (Li and Linde 1999), that is, the probability density function of x_i generally does not exist. But integration over $\mathcal{F}(x_i; R)$ is still well-defined since $f_r(\cdot)$ and $f_y(\cdot)$ are both measurable functions.

3.2. B-Spline Based Sieve M-Estimation

Given the observations $(r_i, \delta_i, \mathbf{z}_i, \mathbf{y}_i)$, $i = 1, \dots, n$, from the joint density function (3) we obtain the log-likelihood for $\boldsymbol{\theta}$ as

$$\ell(\boldsymbol{\theta}) = \sum_{i=1}^n \log \int_{x_i \in \mathcal{X}(\mathcal{T})} f_r(r_i, \delta_i|\mathbf{z}_i, x_i; \boldsymbol{\xi}, \beta, \lambda) \times f_y(\mathbf{y}_i|x_i; R, \mu, \sigma_e^2) d\mathcal{F}(x_i; R), \quad (4)$$

where β, λ, R , and μ are indefinite-dimensional parameters. In this section, we propose a B-spline based sieve M-estimation for the estimation of $\boldsymbol{\theta}$.

Let $\mathcal{K}_{\beta,n} = \{t_k, k = 1, \dots, K_{\beta,n}\}$ be a set of partition points of $\mathcal{T}$ with $K_{\beta,n} = O(n^\nu)$ and $\max_k |t_k - t_{k-1}| = O(n^{-\nu})$ for some constant $\nu \in (0, 1/2)$. Let $\mathcal{S}(\mathcal{K}_{\beta,n}, p)$ denote the space of polynomial splines of order $p \geq 1$ with the knots sequence $\mathcal{K}_{\beta,n}$ as defined in Definition 4.1 of Schumaker (1981). According to Corollary 4.10 of Schumaker (1981), we approximate $\mu(t)$ and $x_i(t)$ by $\mu_n(t)$ and $x_{i,n}(t) \in \mathcal{S}(\mathcal{K}_{\beta,n}, p)$,

$$\mu(t) \simeq \mu_n(t) = \sum_{k=1}^q B_{n,k}(t) u_{n,k} = \mathbf{B}_n(t)^\top \mathbf{u}_n,$$

$$x_i(t) \simeq x_{i,n}(t) = \sum_{k=1}^q B_{n,k}(t) u_{n,k}^{(i)} = \mathbf{B}_n(t)^\top \mathbf{u}_n^{(i)}, \quad i = 1, \dots, n,$$

for $t \in \mathcal{T}$, where $\mathbf{B}_n(t) = [B_{n,1}(t), \dots, B_{n,q}(t)]^\top$ is the vector of B-spline basis for $\mathcal{S}(\mathcal{K}_{\beta,n}, p)$, q is determined by $\mathcal{K}_{\beta,n}$ and p ,

and $\mathbf{u}_n^{(i)} = [u_{n,1}^{(i)}, \dots, u_{n,q}^{(i)}]^\top$ for $i = 0, 1, \dots, n$. Accordingly, the coefficient function $\beta(t)$ is approximated by

$$\beta_n(t) = \sum_{k=1}^q B_{n,k}(t) b_{n,k} = \mathbf{B}_n(t)^\top \mathbf{b}_n,$$

where $\mathbf{b}_n = [b_{n,1}, \dots, b_{n,q}]^\top$. We assume the $\mathbf{u}_n^{(i)}$ are independently and identically distributed multivariate normal $N(0, \Sigma_n)$, independent of the measurement errors e_{ij} . Let $\mathcal{S} \otimes \mathcal{S}(\mathcal{K}_{\beta,n}, p)$ be the space of the tensor-product splines of two variables based on $\mathcal{S}(\mathcal{K}_{\beta,n}, p)$. Then essentially the covariance kernel function $R(\cdot, \cdot)$ is approximated by

$$R_n(s, t) = \text{cov}(x_{i,n}(s), x_{i,n}(t)) = \mathbf{B}_n(s)^\top \Sigma_n \mathbf{B}_n(t),$$

which is a tensor-product spline in $\mathcal{S} \otimes \mathcal{S}(\mathcal{K}_{\beta,n}, p)$.

For the baseline hazard function $\lambda(\varepsilon)$ or equivalently the cumulative hazard $\Lambda(\varepsilon)$, it is well known that the nonparametric maximum likelihood estimation based on a step-function $\Lambda\{\varepsilon\}$ fails for AFT models (Zeng and Lin 2007). To facilitate an efficient estimation, we need to model $\lambda(\varepsilon)$ as a smooth function. To ensure the positivity of $\lambda(\varepsilon)$, we propose to approximate $g(\varepsilon) = \log \lambda(\varepsilon)$ by another B-spline expansion as introduced by Ding and Nan (2011). Specifically, let a and b be prespecified boundaries of the residuals ε_i , where $-\infty < a < b < \infty$. Let $\mathcal{K}_{g,n} = \{\kappa_j, j = 1, \dots, K_{g,n}\}$ be a set of partition points of $[a, b]$ with $K_{g,n} = O(n^{\tilde{\nu}})$ and $\max_j |\kappa_j - \kappa_{j-1}| = O(n^{-\tilde{\nu}})$ for some constant $\tilde{\nu} \in (0, 1/2)$. Let $\mathcal{S}(\mathcal{K}_{g,n}, \tilde{p})$ denote the space of polynomial splines of order $\tilde{p} \geq 1$ with the knots sequence $\mathcal{K}_{g,n}$. Under suitable smoothness assumptions satisfied by most reasonable distributions of ε_i , we can approximate $g(\varepsilon)$ by

$$g_n(\varepsilon) = \sum_{k=1}^{\tilde{q}} \phi_{n,k}(\varepsilon) \gamma_{n,k} = \Phi_n(\varepsilon)^\top \gamma_n,$$

where $\gamma_n = [\gamma_{n,1}, \dots, \gamma_{n,\tilde{q}}]^\top$ and $\Phi_n(\varepsilon) = [\phi_{n,1}(\varepsilon), \dots, \phi_{n,\tilde{q}}(\varepsilon)]^\top$ is the vector of B-spline bases for $\mathcal{S}(\mathcal{K}_{g,n}, \tilde{p})$. The B-spline approximation g_n imposes a semiparametric distribution on ε_i , for which $\lambda(\varepsilon)$ and $\Lambda(\varepsilon)$ are approximated by $\lambda_n(\varepsilon) = \exp\{\Phi_n(\varepsilon)^\top \gamma_n\}$ and $\Lambda_n(\varepsilon) = \int_a^b I(\varepsilon \geq s) \lambda_n(s) ds = \int_a^b I(\varepsilon \geq s) \exp\{\Phi_n(s)^\top \gamma_n\} ds$, respectively.

Let Ξ denote the parameter space of ξ and σ_e^2 . The B-spline based sieve M-estimator of θ is defined as

$$\hat{\theta}_n = (\hat{\xi}, \hat{\beta}_n, \hat{g}_n, \hat{R}_n, \hat{\mu}_n, \hat{\sigma}_e^2) = \arg \max_{\theta_n = (\xi, \beta_n, g_n, R_n, \mu_n, \sigma_e^2)} \ell(\theta_n),$$

where $(\xi, \sigma_e^2) \in \Xi$, $\beta_n, \mu_n \in \mathcal{S}(\mathcal{K}_{\beta,n}, p)$, $g_n \in \mathcal{S}(\mathcal{K}_{g,n}, \tilde{p})$, and $R_n \in \mathcal{S} \otimes \mathcal{S}(\mathcal{K}_{\beta,n}, p)$. Let

$$\tilde{\theta}_n = (\hat{\xi}, \mathbf{b}_n, \gamma_n, \Sigma_n, \mathbf{u}_n, \hat{\sigma}_e^2)$$

which is the Euclidean representation of θ_n . It follows from direct calculations that

$$\begin{aligned} \ell(\theta_n) &= \ell_n(\tilde{\theta}_n) = \sum_{i=1}^n \log \int_{\mathbf{u}_n^{(i)}} f_{r,n}(r_i, \delta_i | \mathbf{u}_i, \mathbf{z}_i; \xi, \mathbf{b}_n, \gamma_n) \\ &\times \left\{ \prod_{j=1}^{m_i} f_{y,n}(y_{ij} | t_{ij}, \mathbf{u}_n^{(i)}; \mathbf{u}_n, \sigma_e^2) \right\} f_u(\mathbf{u}_n^{(i)}; \Sigma_n) d\mathbf{u}_n^{(i)}, \quad (5) \end{aligned}$$

where $f_{y,n}(y_{ij} | t_{ij}, \mathbf{u}_n^{(i)}; \mathbf{u}_n, \sigma_e^2)$ and $f_u(\mathbf{u}_n^{(i)}; \Sigma_n)$ are the densities of $N(\mathbf{B}_n(t_{ij})^\top (\mathbf{u}_n + \mathbf{u}_n^{(i)}), \sigma_e^2)$ and $N(0, \Sigma_n)$. The sieve density function $f_{r,n}(\cdot)$ for the survival data (r_i, δ_i) is given as

$$\begin{aligned} f_{r,n}(r_i, \delta_i | \mathbf{u}_n^{(i)}, \mathbf{z}_i; \xi, \mathbf{b}_n, \gamma_n) \\ = \exp \left\{ \delta_i \Phi_n(r_i - \xi^\top \mathbf{z}_i - \mathbf{b}_n^\top \mathbf{A}_n \mathbf{u}_n^{(i)})^\top \gamma_n \right. \\ \left. - \int_a^b I(r_i - \xi^\top \mathbf{z}_i - \mathbf{b}_n^\top \mathbf{A}_n \mathbf{u}_n^{(i)} \geq s) \exp\{\Phi_n(s)^\top \gamma_n\} ds \right\}, \end{aligned}$$

where $\mathbf{A}_n = \int_{\mathcal{T}} \mathbf{B}_n(t) \mathbf{B}_n(t)^\top dt$.

Similar to the problem confronted in the reduced rank estimator in James, Hastie, and Sugar (2000), maximizing (5) directly is a difficult non-convex optimization problem. To avoid this problem, we take advantage of the B-spline approximations and propose an EM algorithm (Dempster, Laird, and Rubin 1977). More details are provided in the supplementary material.

Remark 4. In practical applications, we need to specify the number and locations of the knots for the B-splines in both $\mathbf{B}(t)$ and $\Phi(\varepsilon)$. The number of knots for $\mathbf{B}(t)$ can be chosen by a preliminary analysis of the biomarker data, in a similar way as Rice and Wu (2001), such as the BIC method. As to the knots locations of $\mathbf{B}(t)$, our limited experiences show that equal-spaced sample quantile knots generally work well. More details on this issue can be found in Wu and Zhang (2006). For $\Phi(\varepsilon)$, using one interior knot located at 0 is usually enough to provide good approximations for a wide range of distributions.

4. Asymptotic Theories

In this section we derive asymptotic theories of the joint model. For ease of exposition, the results do not include the population average μ of the model (1), but similar results hold for its estimator. Without loss of generality, we assume $\mu(t) \equiv 0$. The parameter of interest is now denoted by $\theta = (\xi, \sigma_e^2, \beta(\cdot), g(\cdot), R(\cdot, \cdot))$. The sieve estimator $\hat{\theta}_n$ is defined as $\hat{\theta}_n = (\hat{\xi}, \hat{\sigma}_e^2, \hat{\beta}(\cdot), \hat{g}(\cdot), \hat{R}(\cdot, \cdot)) = (\hat{\xi}, \hat{\sigma}_e^2, \mathbf{B}(\cdot)^\top \hat{\mathbf{b}}, \Phi(\cdot)^\top \hat{\gamma}, \mathbf{B}(\cdot)^\top \hat{\Sigma} \mathbf{B}(\cdot))$.

4.1. Consistency and Efficiency

Let $\psi = (\xi, \sigma_e^2)$. The following theorem states the convergence rate of the sieve estimator $\hat{\theta}_n$ to the true parameter $\theta_0 = (\xi_0, \sigma_0^2, \beta_0(\cdot), g_0(\cdot), R_0(\cdot, \cdot))$.

Theorem 4.1. Let $K_g = O(n^{\tilde{\nu}})$ and $K_\beta = O(n^\nu)$, where $1/(2+2\tilde{p}) < \tilde{\nu} < 1/(2\tilde{p})$ and $1/(4p) < \nu < 1/(2p)$. Suppose the Conditions (C.1)–(C.9) provided in the supplementary material hold and the observations are independent and identically distributed as defined by model (1) and (2). Then

$$d(\hat{\theta}_n, \theta_0) = O_p(n^{-\min\{\tilde{p}\tilde{\nu}, p\nu, (1-\max\{\tilde{\nu}, 2\nu\})/2\}}),$$

where $d(\cdot, \cdot)$ is a distance between $\hat{\theta}$ and θ_0 defined in the supplementary material, and $\tilde{p}$ and p are respectively the smoothness orders of $g(\cdot)$ and $\beta(\cdot)$.

The proof of [Theorem 4.1](#) is given in the supplementary material. In most cases $g(\cdot)$ can be assumed to be smoother than $\beta(\cdot)$. Then the optimal convergence rate implied by [Theorem 4.1](#) is $d(\hat{\theta}_n, \theta_0) = O_p(n^{-p/(2+2p)})$ when $\nu = 1/(2+2p)$, which is dominated by the rate of $\hat{R}(t, s) = \mathbf{B}(t)^\top \hat{\Sigma} \mathbf{B}(s)$. It is the benchmark rate for nonparametric estimation of the bivariate function $R(t, s)$. The next theorem states that optimal rates can be achieved respectively. In particular, the parametric part $\hat{\xi}$ is still $\sqrt{n}$ -consistent and achieves semiparametric efficiency.

Theorem 4.2. Under the same conditions as [Theorem 4.1](#) and suppose $(\hat{p} - 2)\bar{\nu} > \nu$, we have,

$$n^{1/2}(\hat{\xi} - \xi_0) \xrightarrow{d} N(0, \Omega_\xi),$$

where $\xrightarrow{d}$ denotes weak convergence by distribution, and Ω_ξ is a constant positive-definite matrix which achieves the semiparametric efficiency bound for ξ_0 . Moreover, there exists $\beta_{n,0} \in \mathcal{S}(\mathcal{K}_{\beta,n}, p)$, such that $\|\beta_{n,0} - \beta_0\|_{R_0} = O_p(K_{\beta,n}^{-p})$, $\|\hat{\beta} - \beta_{n,0}\|_{R_0} = O_p\{(K_{\beta,n}/n)^{1/2}\}$, where $\|\beta\|_{R_0}^2 = \int_{\mathcal{T}} \int_{\mathcal{T}} \beta(t) R_0(t, s) \beta(s) dt ds$, and

$$(\text{var}\{\hat{\beta}(t)\})^{-1/2}(\hat{\beta}(t) - \beta_{n,0}(t)) \xrightarrow{d} N(0, 1),$$

where $\text{var}\{\hat{\beta}(t)\} \leq \rho n^{-1} \mathbf{B}(t)^\top \left[\int_{\mathcal{T}} \int_{\mathcal{T}} \mathbf{B}(\nu) R_0(\nu, s) \mathbf{B}(s)^\top d\nu ds \right]^{-1} \mathbf{B}(t)$ for a constant ρ .

[Theorem 4.2](#) implies that the optimal convergence rate for $\hat{\beta}$ is $\|\hat{\beta} - \beta_0\|_{R_0} = O_p\{n^{-p/(1+2p)}\}$ when $K_{\beta,n} = O(n^{1/(1+2p)})$, which is the benchmark global rate of convergence for one-dimensional nonparametric estimation. We can interpret $\beta_{n,0}$ as the estimable part of β_0 in the sieve space $\mathcal{S}(\mathcal{K}_{\beta,n}, p)$ (Huang, Wu, and Zhou 2004). The proof of [Theorem 4.2](#) is given in the supplementary material.

4.2. Confidence Intervals

Since model (1) is a typical mixed effects model, standard methods (e.g., Ruppert, Wand, and Carroll 2003; Wu and Zhang 2006) are available to make inferences on $\mu(t)$ and $x_i(t)$. Here, we discuss the inferences for $\beta(t)$ and ξ . The bootstrap approach can be applied here, which was studied by Tseng, Hsieh, and Wang (2005) for the time-dependent AFT model joint with a longitudinal mixed-effects model. But it may be time-consuming for our model due to the repeated use of the EM algorithm.

An alternative approach proposed by Ding and Nan (2011) is to invert the observed information matrix from the last iteration. We propose a similar method based on the EM algorithm. Let $\hat{E}_i[\cdot]$ denote the expectation used in the last iteration of the EM algorithm. The observed Fisher information matrix for all parameters can be estimated by (see, e.g., Ruud 1991)

$$\hat{\mathbf{J}} = \sum_{i=1}^n \left\{ \hat{E}_i \left[\frac{\partial \ell^*(\tilde{\theta}; \mathbf{u}_i)}{\partial \tilde{\theta}} \right] \hat{E}_i \left[\frac{\partial \ell^*(\tilde{\theta}; \mathbf{u}_i)}{\partial \tilde{\theta}} \right]^\top \right\} \Big|_{\tilde{\theta} = \hat{\theta}}.$$

The matrix $\hat{\mathbf{J}}$ is by definition positive-definite, that is, $\hat{\mathbf{J}}^{-1}$ is a well-defined covariance matrix. This estimator is preferred over the estimator proposed by Louis (1982), since Louis's estimator

does not guarantee a positive-definite matrix in our setting, which may lead to negative standard error estimates, and have inaccurate inverses.

The confidence intervals for $\hat{\mathbf{b}}$ and $\hat{\xi}$ can be obtained by extracting the corresponding rows and columns of $\hat{\mathbf{J}}^{-1}$, and resorting to the normal distribution quantiles as suggested by [Theorem 4.2](#). Specifically, denote the variance matrix of $\hat{\mathbf{b}}$ by $\Sigma_{\mathbf{b}}$. Then for any time $t \in \mathcal{T}$, we have $\text{var}\{\hat{\beta}(t)\} = \mathbf{B}(t)^\top \Sigma_{\mathbf{b}} \mathbf{B}(t)$. The pointwise 95% confidence interval for $\hat{\beta}(t)$ can be constructed as $\hat{\beta}(t) \pm 1.96\{\mathbf{B}(t)^\top \Sigma_{\mathbf{b}} \mathbf{B}(t)\}^{1/2}$.

4.3. Failure Time Ratio

Based on the semiparametric joint model, the conditional mean of the log-transformed survival time can be easily obtained as

$$\begin{aligned} E\{\log(T_i) | \mathbf{z}_i, \mathbf{y}_i\} &= \xi^\top \mathbf{z}_i + \int_{\mathcal{T}} \beta(t) E\{x_i(t) | \mathbf{y}_i\} dt \\ &= \xi^\top \mathbf{z}_i + \int_{\mathcal{T}} \beta(t) R(t, \mathbf{t}_i) dt \\ &\quad \times (R(\mathbf{t}_i, \mathbf{t}_i) + \sigma_e^2 I_{m_i})^{-1} \mathbf{y}_i, \end{aligned} \quad (6)$$

which is a linear function of both $\mathbf{z}_i$ and $\mathbf{y}_i$. Therefore, model (2) enjoys a straightforward interpretation in the same way as the classic semiparametric linear model (see, e.g., Wei 1992; Ding and Nan 2011). Specifically, the baseline coefficients $\exp\{\xi_k\}$, $k = 1, \dots, d$, can be interpreted as the *Failure Time Ratio* (FTR): a unit change on z_{ik} , keeping all other covariates fixed, leads to a proportional change of the survival time by the factor $\exp\{\xi_k\}$.

5. Simulation Studies

Comparing the proposed model with the existing semiparametric time-varying survival models is nontrivial because the coefficients produced by each model have different meanings and are not comparable. Moreover, a key utility of semiparametric joint model is to estimate the survival times, which is not feasible in most semiparametric time-varying models. To facilitate a comparison, we considered a parametric time-varying proportional hazard (PH) model with Weibull baseline hazard, where the longitudinal trajectory is treated as a time-varying covariate. Because estimating the survival time in a time-varying model requires extrapolating the longitudinal trajectory beyond the observational window, we employed a parametric cubic polynomial model for the extrapolation. The parametric PH model and the cubic polynomial model are estimated simultaneously via the classical joint modeling approach, using the JM package (Rizopoulos 2010). For the time-varying PH model, we propose to estimate the mean survival time by $\hat{E}_{\text{PH}}[T | \mathbf{z}_i, \mathbf{y}_i] = \int_0^\infty \hat{S}_{\text{PH}}(t | \mathbf{z}_i, \mathbf{y}_i) dt$, where $\hat{S}_{\text{PH}}(t | \mathbf{z}_i, \mathbf{y}_i)$ is the estimated conditional survival function of the parametric PH model. The above integral does not have a closed form which is estimated by numerical integration. This is compared with our semiparametric joint model with the survival time estimated by methods described in [Section 2.2](#).

We designed three simulation examples with a range of parameter settings to evaluate the finite sample performance

of the proposed semiparametric joint model, generating 100 replicates of data for each setting. Due to space limit, we present the first two simulations here and include the third simulation study in the supplementary material. The accuracy of the estimated survival times was assessed by the standardized bias and mean squared error (MSE) with respect to the underlying “true” mean survival times $\bar{T}_i$. Specifically, $\text{Bias}_T = n^{-1} \sum_{i=1}^n (\hat{T}_i - \bar{T}_i) / \text{std}(\bar{T}_i)$ and $\text{MSE}_T = n^{-1} \sum_{i=1}^n (\hat{T}_i - \bar{T}_i)^2 / \text{var}(\bar{T}_i)$.

5.1. Example 1

In the first example we generated the survival data by the model: $\log T_i = 2 + z_{i,1}/2 + z_{i,2}/2 + \int_{\mathcal{T}} \beta(t)x_i(t)dt + \varepsilon_i$, where $\mathcal{T} = [-25, 100]$, $\beta(t) = -3/100 + 10(t+25)/100^2 - 6(t+25)/100^3 - 0.5(t+25)/100^4$, and $\exp\{\varepsilon_i\}$ follows a Weibull distribution with scale = 2 and shape = 5. The time-independent covariates were generated by $z_{i,1} \sim N(0, 0.5^2)$ and $z_{i,2} \sim \text{Bin}(0.5)$. With an additional functional term, this model is an extension of a popular simulation model considered in AFT literature, see, for example, Zeng and Lin (2007) and Ding and Nan (2011). Although the magnitude of the functional coefficient $\beta(t)$ looks small, the variance of the integral term is roughly the same as the variance of the other two covariate terms in this model. The censoring times C_i were generated from a uniform $[0, \tau]$ distribution, where τ was chosen to produce a given censoring rate. The observed survival data were $\{S_i = \min\{T_i, C_i\}, \delta_i = I(T_i \leq C_i)\}$. The longitudinal trajectory was assumed to follow a cubic polynomial function: $y_i(t) = \mu(t) + x_i(t) + e_i(t)$, where $\mu(t) = 3.5 - t/100 + (t/100)^2 - 8(t/100)^3$, $x_i(t) = u_{i0}/10 + u_{i1}t/100 + u_{i2}(t/100)^2 + u_{i3}(t/100)^3$, $e_i(t) \sim N(0, \sigma_e^2)$, and $\mathbf{u}_i = (u_{i0}, u_{i1}, u_{i2}, u_{i3})^\top \sim N(0, \Sigma_u)$ where $\Sigma_u = \text{diag}(1, 2, 4, 6)$. The longitudinal data were sampled following a similar design as the AASK data considered in Section 6. Specifically, for the i th subject, we first drew a time t_{i0} from the uniform distribution on $[-25, -15]$ to represent a random starting point before $t = 0$. Then we generated observation time points $\{t_{i0}, t_{i0} + 5 + \tilde{t}_{i1}, \dots, t_{i0} + 125 + \tilde{t}_{i25}\}$, where $\tilde{t}_{ik} \sim_{\text{iid}} N(0, 1)$ to create a non-equidistant “jittered” design for subject-specific samples. Finally, only the longitudinal data within the interval $[-25, S_i]$ were included in the analytic sample.

Eight different simulation settings were conducted varying: (a) the sample size $n = 200$ and 400 ; (b) the level of measurement errors for the covariate data: $\sigma_e = 0.05$ and 0.15 , that is, signal to noise ratio was approximately 12 and 4, respectively; and (c) the censor rate CR = 10%, 40%. Two estimation methods were implemented: (a) parametric time-varying proportional hazard model joint with a cubic polynomial longitudinal model, where the baseline hazard is modeled by the Weibull distribution; (b) the proposed semiparametric joint model based on cubic B-spline approximations of $\beta(t)$, $\mu(t)$ and $x_i(t)$, where we used BIC to select the optimal number of knots as suggested by Rice and Wu (2001). The log-baseline hazard $g(t) = \log \lambda(t)$ was estimated by a cubic B-spline on $[-3, 3]$ with one interior knot at 0. Initial sample size for Monte Carlo integration was $M = 1000$. On average, it took about 5 min (range 1–10 min) to compute our model for sample size = 200; and about 12 min (range 6–20 min) for sample size = 400.

The optimal number of knots determined by BIC was consistently estimated as 0 in almost all replications and all settings,

which leads to 4 random coefficients that is in accordance with the underlying true cubic polynomial model. Figure 2 summarizes the results for Example 2. Our proposed estimation approach produces unbiased survival time estimation in all settings, while the time-varying PH model suffers a systematic negative bias. The MSE of the semiparametric joint model is much smaller than the time-varying PH model by a clear margin. Combining the results of the two simulation studies, we conclude that our proposed model is a better choice than the time-varying PH model in survival time estimation, in terms of protection against systematic biases and robustness to model misspecification, censor rates, and signal to noise ratios.

The finite-sample performance of our proposed method was further evaluated by the estimation accuracy of the coefficients. Let $\xi = (\xi_1, \xi_2)$ denote the parametric coefficient of $\mathbf{z}_i = (z_{i1}, z_{i2})$. As shown in Table 1, the estimated parametric coefficients $\hat{\xi}$ are virtually unbiased. We can observe a clear error reduction when the sample size increases, showing the consistency and efficiency of our method. The estimation error for the functional coefficient $\beta(t)$ was measured by $\text{MSE}(\hat{\beta}) = \|\hat{\beta} - \beta_0\|_R^2 = \int_0^1 \int_0^1 [\hat{\beta} - \beta_0](s)R(s, t)[\hat{\beta} - \beta_0](t)dsdt$ where $R(s, t) = \text{cov}(x(s), x(t))$, which is widely used for functional linear models (e.g., Qu, Wang, and Wang 2016). Not surprisingly, increasing either the censoring rate or the level of measurement errors in the covariate trajectories reduces the estimation accuracy of β . The consistency of the β estimation is shown by the decreased errors as sample size increases.

In Figure 3, we illustrate the distributional properties of $\hat{\xi}$ and $\hat{\beta}(0)$ estimated in the 100 replications under the setting $\sigma_e = 0.15$ and censor rate 40% by their QQ-plots and histograms. This setting has the least signal-to-noise ratio and thus is the hardest to estimate. The plots of other settings are provided in the supplementary material. We can see that the estimated coefficients largely follow a normal distribution as proved in Theorem 4.2. In addition, we performed three more simulation studies to show that estimate is not sensitive to the number and location of the knots under reasonable choices, and is robust to autocorrelations of the error terms e_{ij} . First, we used one more knot than that selected by BIC. Second, we used equal-spaced interior knots. Third, we generated the error terms e_{ij} by an AR(1) process with autocorrelation = 0.7. The estimates of ξ_1 and ξ_2 in all three scenarios were similar to those in Table 1. The estimates of $\beta(\cdot)$ were slightly worse but still within a reasonable range. More details are given in the supplementary material.

5.2. Example 2

In the second example we performed a sensitivity analysis by generating the survival data by a time-varying PH model. The hazard function was $\lambda(T_i = t | \bar{X}_i(t)) = \lambda_0(t) \exp\{3 + z_{i,1}/2 + z_{i,2}/2 - 6x_i(t)\}$, where $\lambda_0(t) = 2.5(t/2)^4$ which was the same Weibull distribution as Example 1. The censoring times C_i , survival data $\{S_i = \min\{T_i, C_i\}, \delta_i = I(T_i \leq C_i)\}$, covariates $z_{i,1}$, $z_{i,2}$ and the longitudinal data were generated in the same way as Example 1. Similar to Example 1, eight different simulation settings were conducted varying: (a) the sample size $n = 200$ and 400 ; (b) the level of measurement errors: $\sigma_e = 0.05$ and 0.15 ; and (c) the censor rate CR = 10%, 40%; the same two estimation methods were implemented.

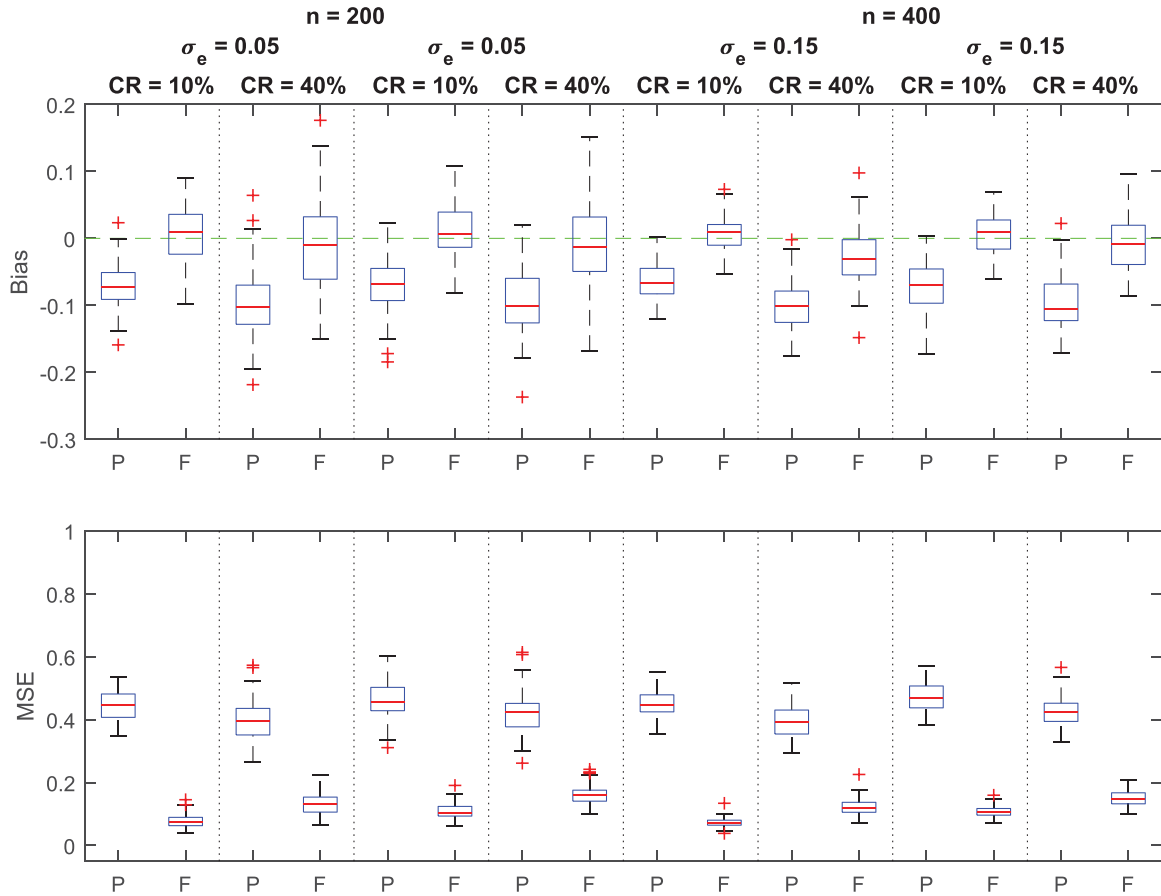


Figure 2. Results of Example 1: data generated by the functional linear model. P: time-varying PH model; F: functional model.

Figure 4 summarizes the results for Example 2. It shows that the survival time estimations under the true PH model specification suffer a systematic positive bias. Such bias was largely due to the unstableness of the longitudinal trajectory extrapolation required for survival time estimation in the time-varying PH model. The positive bias came from the right skewness induced by the exponential transformation of the extrapolated trajectory assumed in the proportional hazard formulation. It implies that even if we estimate the underlying PH model exactly, using such model for survival time prediction would risk overestimating the underlying survival times. In contrast, the functional model was unbiased in more than half of the settings and less biased than the PH model in other settings, even under model misspecification. Moreover, the PH model was shown to be sensitive to increasing measurement errors in the covariate trajectories and/or censor rates. Consequently, in the settings of $\sigma_e = 0.15$ and censor rate = 40%, which was close to the real data collected in the AASK study, the PH model was clearly biased and the MSE was more volatile with wider spread than the functional model.

6. Application

The AASK study was a multicenter randomized clinical trial of 1094 African Americans aged 18–70 years with CKD (Wright et al. 2002). The follow-up period ranges from 3.0 to 12.4 years. Lab and clinical data were collected at baseline, 3 months, 6 months and approximately every 6 months thereafter until ESRD, death or censoring. Estimating time-to-ESRD is important for both patient counseling and the timing of interventions,

Table 1. Estimated coefficients of the functional linear model in the Example 1.

n	σ_e	CR	Bias(ξ_1)	$\sqrt{MSE(\xi_1)}$	Bias(ξ_2)	$\sqrt{MSE(\xi_2)}$	$\sqrt{MSE(\beta)}$
200	0.05	10%	0.00434	0.0329	0.00778	0.0376	0.0446
		40%	0.00453	0.0452	0.0107	0.0433	0.0520
	0.15	10%	0.00393	0.0382	0.00406	0.0404	0.0545
		40%	0.00112	0.0513	0.00644	0.0494	0.0686
400	0.05	10%	0.00146	0.0236	0.00646	0.0245	0.0332
		40%	0.000614	0.0271	0.0112	0.0320	0.0398
	0.15	10%	0.00789	0.0266	0.0159	0.0317	0.0455
		40%	0.00387	0.0283	0.0132	0.0339	0.0489

which is complicated due to the diversity of the CKD progression patterns.

The primary goal of this analysis was to quantify the associations between the eGFR trajectory and the time-to-ESRD after the patient reaches the 30mL/min/1.73m² eGFR threshold. In this analysis, the eGFR trajectories were first aligned using a linear shift such that $t = 0$ when eGFR equals 30mL/min/1.73m². We considered the entire observed trajectory after this point, plus eGFR trajectories in the 3 years prior, similar to Grams et al. (2015). The eGFR values were log-transformed for analysis. Time-to-ESRD was calculated as the months from an eGFR of 30mL/min/1.73m² to ESRD. Both administrative censoring and pre-ESRD death are treated as censoring. Due to discrete visits, $T_{\{eGFR=30\}}$ may not be directly observed. In such cases, we estimated it by a linear interpolation based on the nearest two observations. The time-independent variable $\mathbf{z}_i$ consists of age, sex, urine protein creatinine ratio (UP:Cr), body mass

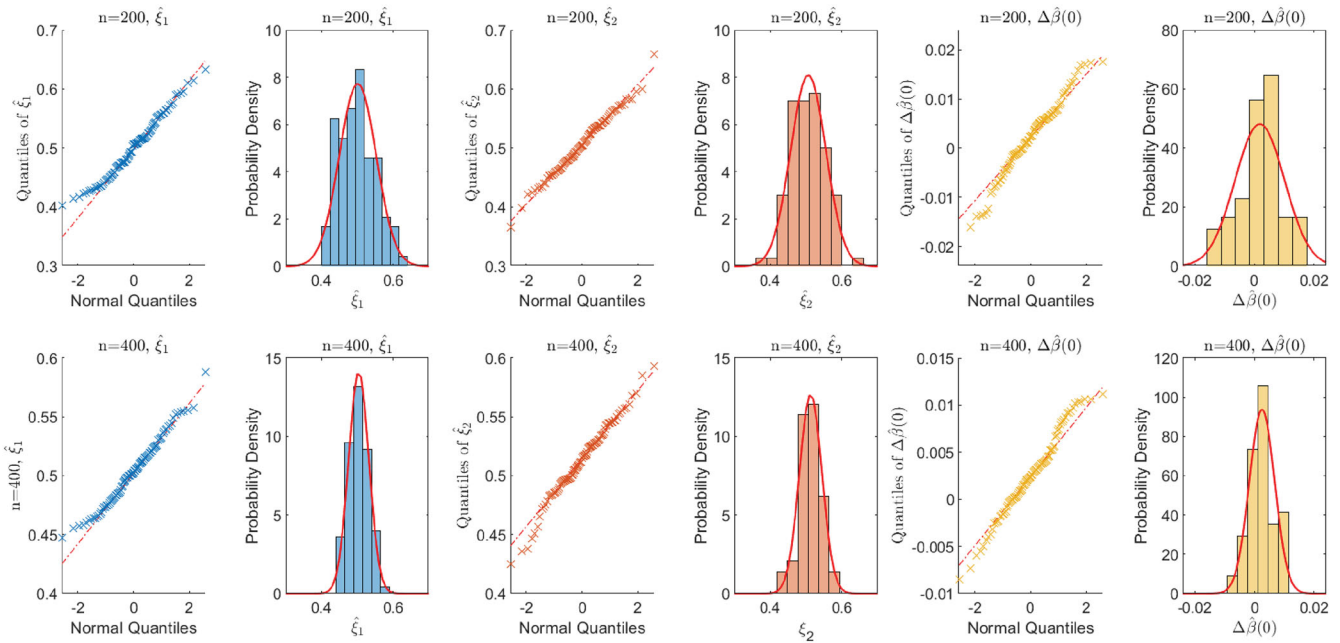


Figure 3. QQ-plots and histograms of the estimated coefficients $\hat{\xi}$ and the estimation error of $\hat{\beta}(t)$ at $t = 0$, denoted by $\Delta\hat{\beta}(0) = \hat{\beta}(0) - \beta(0)$, in the 100 replications under the setting $\sigma_e = 0.15$ and censor rate 40% in the Example 1.

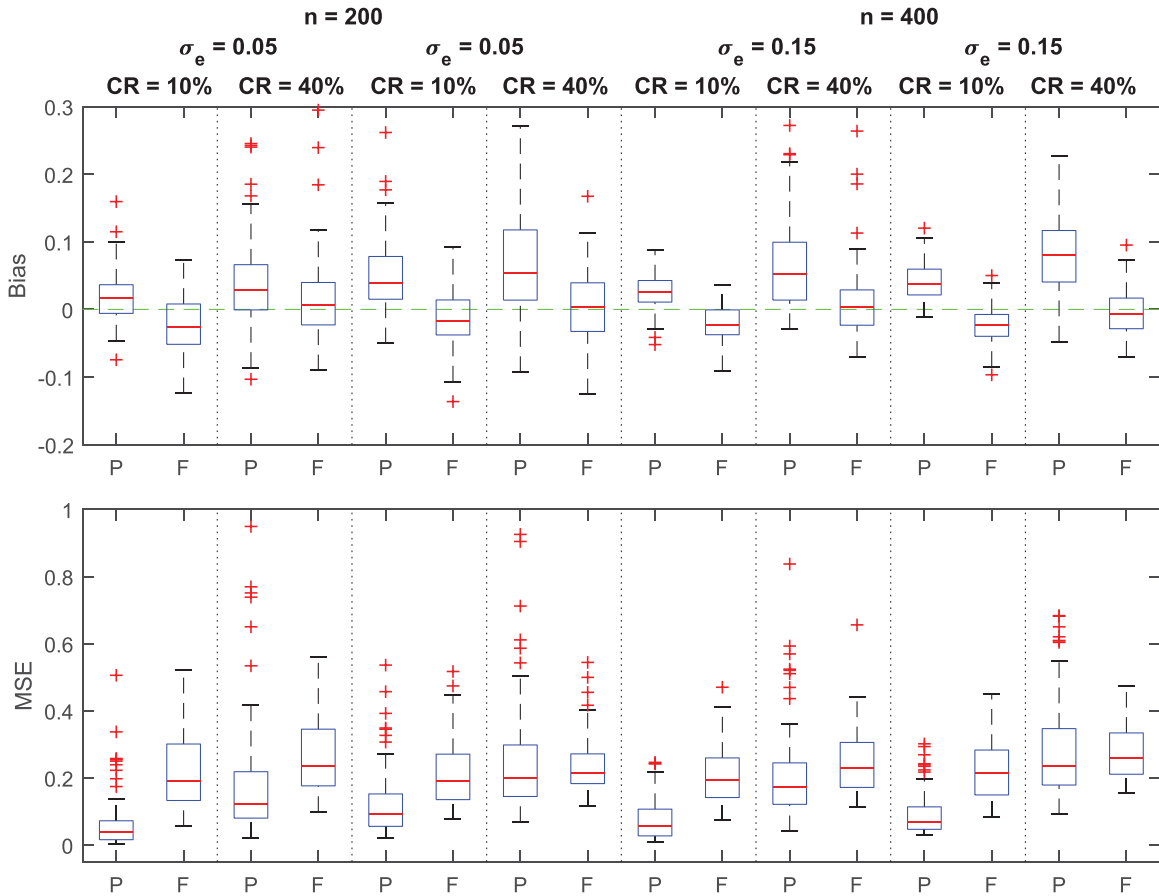


Figure 4. Results of Example 2: data generated by the time-varying PH model. P: time-varying PH model; F: functional model.

index, and systolic blood pressure. In this analysis, we only considered participants who had an eGFR value less than or equal to 30mL/min/1.73m², confirmed by a repeated measurement, and had at least five visits during the study. As a result,

the analytic sample consisted of 313 subjects with 184 (58.8%) reaching ESRD before censoring. To illustrate the connection between eGFR profiles and survival, we divided the subjects into short survivors and long survivors by the median of the survival

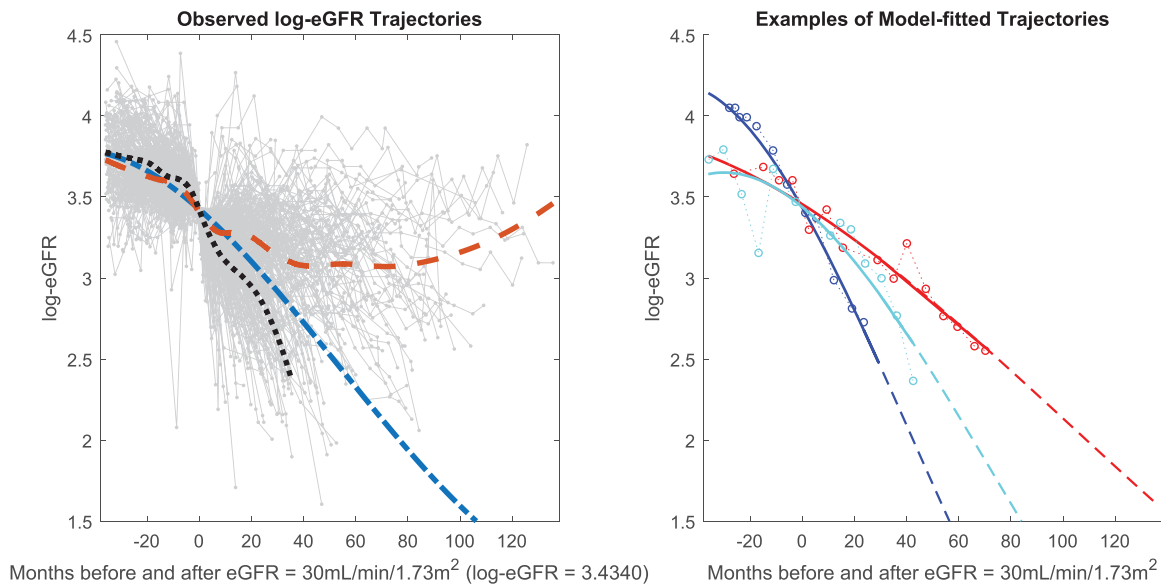


Figure 5. Left panel: spaghetti plots for the log-transformed eGFR values. Dashed line: long survivors; dash-dotted line: short survivors; thick solid line: overall average. Each grey line represents an observed individual eGFR trajectory. Right panel: three examples of fitted individual trajectories.

Table 2. Estimated coefficients of the baseline variables.

	Time-Dep PH Hazard ratio ⁻¹ (95% CI)	Functional Failure time ratio (95% CI)
Age (per 10-y+)	0.782 (0.661 – 0.925)	0.967 (0.920 – 1.016)
Male gender	0.702 (0.502 – 0.983)	0.933 (0.860 – 1.013)
UP:Cr (per 0.1 g/g+)	0.975 (0.962 – 0.989)	0.955 (0.913 – 0.998)
BMI (per 5-kg/m ² +))	0.960 (0.861 – 1.072)	1.012 (0.972 – 1.052)
Systolic BP (per 10-mm Hg+)	0.885 (0.806 – 0.973)	1.014 (0.980 – 1.049)

times. Figure 5 shows the average curves of the short and long survivors.

We analyzed the data using both the time-varying PH model and the proposed semiparametric joint model. For the PH model, to facilitate an estimation of the mean survival time that requires extrapolating the longitudinal trajectory up to infinity, we modeled the longitudinal eGFR profile by a parametric cubic polynomial function. The model was estimated by the JM package (Rizopoulos 2010). For the semiparametric joint model, the longitudinal eGFR profile was modeled by a B-spline expansion, where the interval $\mathcal{T}$ of the B-spline was chosen as the actual span of the pooled observed data, that is $[-36, 138]$ months. We used no interior knot, which was selected by minimizing the BIC scores.

The estimated coefficients of the time-independent variables for both models are shown in Table 2. For the PH model, after controlling for the eGFR profile, the baseline age, gender, MAP and UP:Cr were all significant predictors at level 0.05. Since age and gender were included in the eGFR calculation, that they were still significant indicates that the eGFR information may not be adequately used. For the semiparametric joint model, after controlling for the eGFR profile, only UP:Cr showed a significant association with time to ESRD, at the 0.05 level.

Figure 6(a) shows the estimated coefficient function $\hat{\beta}(\cdot)$; Figure 6(b) shows the estimated latent functional pattern of the three representative longitudinal eGFR trajectories in Figure 1. Their estimated times-to-ESRD are very close to the actually observed. Figure 7 shows the estimated time-to-ESRD of the

whole sample respectively from the time-varying PH model and the semiparametric joint model. Due to the uncertainty of the out-of-sample extrapolation, a considerable number of estimates from the time-varying PH model overestimated the time-to-ESRD by a huge margin as compared to that actually observed. This overestimation issue is very concerning as being too optimistic about the survival time may cause the patient to miss the best timing of intervention. In contrast, the proposed model was much more robust in estimating the time-to-ESRD. As to the censoring times, we purposely ignored the fact that the actual survival times were greater than the censoring times, by not using the truncated distribution. However, the fitted log-survival times still reflect this trait by predicting longer survival times than censoring times for most of the censored subjects.

7. Discussion

We have proposed a semiparametric joint model for time-to-event and longitudinal biomarker by treating the longitudinal trajectories as random realizations of a latent global functional pattern defined in a common conceptual domain. Conditional on the latent functional pattern, the survival part of the model is time independent and inherits the straightforward interpretation of the time independent survival models. The latent functional patterns are estimated from the observed data and integrated out in the overall estimation and inference. Existing functional survival models (Qu, Wang, and Wang 2016; Liu et al. 2025) directly model the observed functional patterns and require the functional data to be fully observed at a common support interval. In contrast, our proposed model considers longitudinal trajectories observed in different time windows that are not directly comparable. Our model only requires the latent pattern to be defined in a common support.

This article extends the scope of joint modeling by using a flexible functional pattern as the shared parameter modeled as a realization of a Gaussian process. Notably, several authors

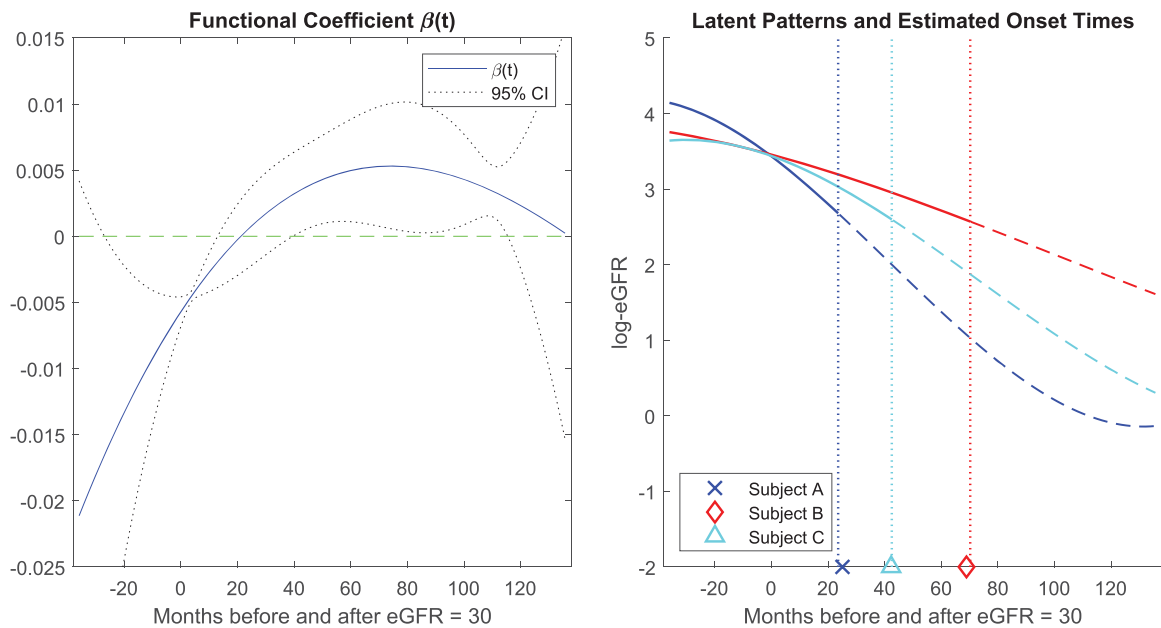


Figure 6. (a) The estimated functional coefficient $\hat{\beta}(t)$. (b) Three representative eGFR patterns corresponding to different lengths of time-to-ESRD. The vertical dotted lines denote the observed time-to-ESRD; the cross/diamond/triangle signs on the x-axis denote the model estimated time-to-ESRD.

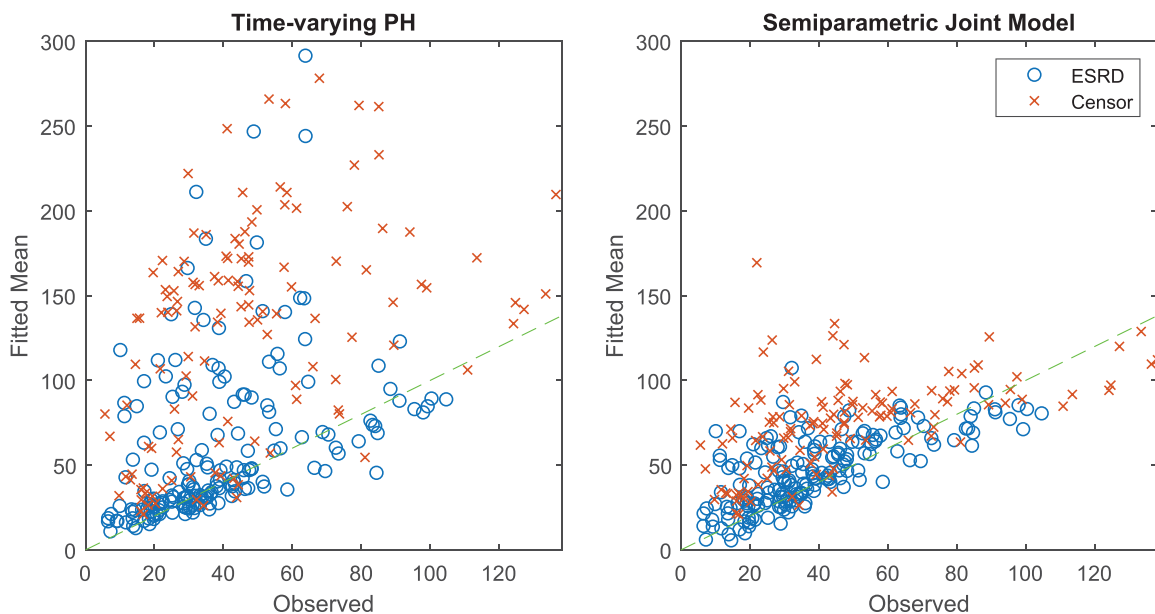


Figure 7. Estimated survival times by the time-varying PH model (left) and the semiparametric joint model (right). 1) Circles: observed time-to-ESRD; 2) Crosses: censored.

have considered using a flexible latent process to model the longitudinal trajectories (Henderson, Diggle, and Dobson 2000; Xu and Zeger 2001; Wang and Taylor 2001). However, their latent process models are designed to impute or extrapolate the longitudinal trajectories. Consequently, the longitudinal value at every single observed survival time needs to be treated as a random effect in joint modeling, leading to high dimensional integration with its dimension growing at the same order as the sample size. To make such models computationally tractable, additional restrictions are usually needed, such as discretizing the survival times to have only a finite number of possible values (Barret et al. 2015). In contrast, the calculation of our joint likelihood function only requires one-dimensional integrations, resulting in a simple computation and rigorous large sample asymptotic theories.

Supplementary Materials

Online supplementary material provides details of the EM algorithm, more simulation results for Example 1, an additional simulation example, and the regularity conditions and proofs for the theorems. Also provided are the computer codes to implement the proposed method and numerical studies.

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The authors report that there are no competing interests to declare.

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